# Do Inert Patterns Explain Sudoku Difficulty?

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Abstract. We ask whether structurally valid but currently inert pattern instances explain variation in human Sudoku solving time. We separate a pattern's carrier condition from its deductive yield, define an inert-at-entry ratio over unique instances, and prove a monotonicity lemma showing that an instance once inert remains inert. The hypothesis is tested on 138 puzzles requiring techniques beyond singles, against average human solving times, under a success criterion fixed in advance. Every pre-specified metric improved in-sample fit and reduced cross-validated fit in the primary model. The hypothesis failed its pre-specified test and is rejected as a predictive account for this corpus.

## 1 Introduction

While solving, one repeatedly encounters structures that satisfy the preconditions of a rule and license nothing. A naked pair sits in a box with both of its digits already eliminated from every other cell in that box. The pattern is present. Recognising it changes nothing about the grid.

Each such structure still costs something to check. The question addressed here is whether that cost is a property of the puzzle, and whether it is one that difficulty ratings fail to record.

Conventional ratings measure the path to the solution. They record the most advanced technique a puzzle requires and the depth of the inference chain leading to the answer. Pelánek's evaluation against human solving data [1] adds a second dimension, the dependency structure among steps, and finds that both matter. Neither records what a solver must discard along the way.

We hypothesise that structurally valid but currently inert pattern instances constitute a *verification burden*, and that their prevalence explains variation in human solving time left unexplained by conventional measures.

Section 2 makes the notion precise and establishes a monotonicity lemma. Section 3 states the experimental design, including the criterion under which the hypothesis would be rejected. Section 4 reports the outcome.

## 2 Formalization

### 2.1 States, instances, and yield.

**Definition 2.1 (*State*).** A *state* $S$ consists of the current cell assignments together with the candidate grid $C(S)$ obtained by eliminating candidates inconsistent with those assignments. No deduction rule is applied during construction of $C(S)$.

**Remark 2.2.** The restriction matters. If singles were applied while constructing the candidate grid, no single would ever be detectable at a state, and the population of patterns would depend on the construction procedure rather than on the grid.

**Definition 2.3** (*Pattern instance and carrier condition*)**.** A *pattern instance* $p$ is a rule together with a location satisfying the rule's *carrier condition*, meaning its structural precondition alone. Write $E(p)$ for the set of candidate eliminations that $p$ licenses. The set $E(p)$ is determined by the rule and the location, and not by the state.

The separation is deliberate. Implementations differ in whether the precondition of a naked pair includes the existence of a candidate to remove. Under that stronger reading the objects studied here do not exist. We therefore fix the weaker structural reading and treat productivity as a separate question.

**Definition 2.4** (*Yield*)**.** The *yield* of $p$ at $S$ is the number of its targets still live,

$$\Delta(p, S) \;=\; \big|E(p) \cap C(S)\big|.$$

We call $p$ *productive* at $S$ when $\Delta(p, S) > 0$, and *inert* at $S$ when $\Delta(p, S) = 0$.

### 2.2 Monotonicity.

**Lemma 2.5** (*Monotonicity*)**.** *Let $p$ satisfy its carrier condition at states $S_t$ and $S_{t'}$ with $t' > t$ along a solve. Then*

$$\Delta(p, S_{t'}) \;\leq\; \Delta(p, S_t).$$

*In particular, $\Delta(p, S_t) = 0$ implies $\Delta(p, S_{t'}) = 0$ for every $t' > t$ at which $p$ persists.*

*Proof.* Candidates are only removed along a solve, so $C(S_{t'}) \subseteq C(S_t)$. A candidate not live at $S_t$ therefore cannot become live at $S_{t'}$. Since $E(p)$ depends on the rule and location alone, it is the same set at both states, and

$$E(p) \cap C(S_{t'}) \;\subseteq\; E(p) \cap C(S_t).$$

Taking cardinalities gives the inequality, and the case $\Delta(p, S_t) = 0$ forces the left side to be empty. ■

Once inert, always inert. Lemma 2.5 justifies recording each instance once at first detection, and it corrects an assumption made in the opposite direction at the outset of this work. The property was also verified computationally across 458 distinct puzzles, with no instance observed to pass from inert to productive.

### 2.3 The metric and what it does not measure.

**Definition 2.6** (*Inert-at-entry ratio*)**.** Let $\mathcal{R}$ be a fixed rule set and let a reference solver generate a path $S_0, \ldots, S_T$. Record each distinct instance once, at the state where it first becomes detectable, and classify it by its yield there. Then

$$\mathrm{IER} \;=\; \frac{\#\{\text{unique instances inert at first detection}\}}{\#\{\text{unique instances detected at any point}\}}.$$

Singles are excluded from $\mathcal{R}$. A detectable single is almost always productive, so their inclusion would make the ratio track how many singles a puzzle offers rather than how many inert structures it presents.

By Lemma 2.5 an instance inert at entry remains inert. An instance productive at entry may later become inert, and is not counted in the numerator. The quantity is therefore a proportion evaluated at entry and not an absolute abundance of inert structure. We use the narrower reading throughout.

Three further specifications are reported as robustness checks. Writing $u(S)$ and $k(S)$ for the counts of inert and productive instances detectable at $S$:

- *Exposure-weighted ratio*, $\sum_t u(S_t) / \sum_t \big(u(S_t) + k(S_t)\big)$, which recounts a persistent inert structure at every state where it remains detectable, and so assumes a solver that re-examines without memory.
- *Raw inert count*, the numerator of IER taken alone, without normalisation by the size of the detectable population.
- *Expected random-order entry cost*, the mean over path states of $u(S)/(k(S)+1)$. If instances at $S$ are examined in uniformly random order, each inert instance precedes all $k(S)$ productive ones with probability $1/(k(S)+1)$, so this is the expected number of inert instances examined before the first productive one.

## 3 Experimental Design

**3.1 Corpus and its relation to development.** The reference solver and the metrics were developed against a set of 344 puzzles carrying human difficulty scores. Those puzzles were used for implementation validation only, including the computational check of Lemma 2.5, and not for selecting the success criterion.

The test corpus consists of 138 puzzles drawn from a public dataset of 1533 puzzles carrying average human solving times [2], retained when naked and hidden singles alone do not solve them, so that every retained puzzle requires at least one rule from the counted population. The two sets are not disjoint: 24 of the 138 test puzzles, or 17 percent, also appear in the development set. The test corpus is therefore not a clean holdout, and "pre-specified" should be read as applying to the analysis plan rather than to the sample.

The dependent variable is the natural logarithm of mean solving time in seconds. All reported correlations are against that transformed variable.

**3.2 Statistical procedure.** Models are ordinary least squares. Controls are clue count, productive step count, mean candidate count, highest required technique, and the mean number of productive instances available per state. Highest required technique is encoded ordinally on the reference solver's three levels. Predictors are not standardised, which affects coefficient magnitudes but not fit statistics.

Cross-validation uses five folds with shuffling. Fold assignments are regenerated under twenty random seeds, numbered 0 through 19, and the reported cross-validated $R^2$ is the mean across all folds and seeds. Within each fold, $R^2$ is computed as $1 - \mathrm{SSE}/\mathrm{SST}$ on the held-out observations, with SST taken about the training-fold mean. Sign stability is assessed on training-fold coefficients and reported as the number of the twenty fold assignments under which the sign is retained in all five folds.

**3.3 Criteria fixed before the test.**

**Criterion 3.1 (*Adequacy*).** At least one control predictor must reach $|r| \geq 0.25$ against log mean solving time, the level reported for the weakest published baseline in [1].

Absent this, a null result would be uninformative about the hypothesis and informative only about the corpus.

**Criterion 3.2 (*Success*).** The metric must improve cross-validated $R^2$ over the control model, with a coefficient holding its sign across folds. Improvement in in-sample $R^2$ does not count, since adding any variable produces it.

## 4 Results

**4.1 Adequacy, and one control that must be discarded.** Three controls clear the adequacy threshold on their face: productive step count at $r = -0.332$, mean candidate count at $+0.282$, and highest required technique at $+0.258$.

The first is an artefact and is discarded as evidence for adequacy, although it remains in the pre-specified control model reported below and in Table 1. Productive step count correlates with solver completion at $r = +0.876$, so it is close to a restatement of whether the reference solver finished. Completed puzzles average 53.3 recorded steps and stalled puzzles 26.3. Within the completed subset alone the correlation between step count and solving time falls to $+0.067$. The negative sign therefore reflects truncation, since harder puzzles stall the solver earlier and contribute shorter observed paths, and not any property of the puzzles themselves.

Mean candidate count and highest required technique survive, so the adequacy criterion is met independently of the contaminated predictor. The control model, unchanged from the plan, attains cross-validated $R^2 = +0.152$.

| Added metric | In-sample $\Delta R^2$ | Cross-validated $\Delta R^2$ |
|---|---|---|
| IER (primary) | $+0.0070$ | $-0.0067$ |
| Exposure-weighted ratio | $+0.0002$ | $-0.0236$ |
| Raw inert count | $+0.0030$ | $-0.0167$ |
| Expected random-order entry cost | $+0.0117$ | $-0.0040$ |

Table 1. Incremental value over the control model, $n = 138$.

Every pre-specified specification slightly improved in-sample fit and reduced out-of-sample performance. Three alternative specifications produce the same conclusion as the primary one.

The second half of the success criterion behaves differently and is reported here in corrected form. The IER coefficient retains its sign across all five folds under 17 of the 20 fold assignments. It is therefore reasonably stable, and the failure is located in predictive value rather than in instability.

**4.2 Two specifications outside the pre-specified plan.** Because productive step count proved to be a truncation artefact, we refit the control model with a solver-completion indicator added. Control performance improves to cross-validated $R^2 = +0.1815$, and adding IER gives $+0.1824$, a change of $+0.0009$. The sign of the increment reverses relative to the pre-specified specification, but its magnitude is under one tenth of one percent of variance.

Restricting to the 56 fully solved puzzles, the control model attains cross-validated

$R^2 = -0.1998$, so it does not generalise at that sample size, and adding IER moves it to $-0.2601$. The correlation between IER and solving time in that subset is $+0.119$. No conclusion rests on this subset.

Neither specification was fixed in advance and neither is treated as a test.

**Outcome.** The first half of the success criterion is not met. The hypothesis failed its pre-specified test and is rejected as a predictive account for this corpus.

## 5 Scope of the Conclusion

The claim is bounded in four ways.

First, the study finds no out-of-sample evidence that inert-pattern prevalence improves difficulty prediction here. It does not establish that no such relation exists. With 138 puzzles and a median of 12 players per puzzle, the design has limited power against a small effect.

Second, and more importantly, the metric counts structures a reference solver can enumerate. It does not count structures a human inspects. These are different quantities, and the study measures only the first. A human solver plausibly never looks at most of the instances counted here. The null result may therefore reflect a failure of the proxy rather than the absence of verification cost itself.

Third, the result is sensitive to the control specification. Under the pre-specified controls the increment is negative, and under a post-hoc model controlling for solver completion it is positive by $+0.0009$. A quantity that changes sign under a defensible respecification of the controls is not established in either direction. Across the specifications examined, the largest observed mean increment was $+0.0009$.

Fourth, the asymmetry expected from [1], whose dependency metric counts productive options, is not reproduced. Mean productive availability reaches only $r = +0.132$ in this corpus and does not clear the adequacy threshold, so the comparison cannot be made here.

## 6 Limitations

The median number of players per puzzle is 12. The datasets used in [1] have 131 and 1307. Measurement error in the dependent variable attenuates every association reported here, including those of the controls.

The counted population covers locked candidates, naked pairs, and hidden pairs. Triples, fish patterns, and chains are absent.

The reference solver fully solves 56 of the 138 puzzles. For the remaining 82 the metrics are computed over a truncated path, and puzzles that stall the solver earlier are observed over shorter paths. This is the most serious defect in the design. It disqualified one control outright, and it means every path-aggregated quantity here is observed unevenly across the corpus.

Highest required technique is defined by the reference solver and is therefore not independent of the instrumentation.

One corpus, one platform, with the development overlap noted in Section 3.

## 7 Research Notes

The hypothesis began with repeated encounters with logically valid patterns that produced no elimination. Rather than treating that observation as evidence, it was treated as a hypothesis, and the result that would falsify it was specified in advance.

The hypothesis failed its pre-specified test, and it was rejected.

## References

[1] R. Pelánek, *Difficulty rating of Sudoku puzzles: an overview and evaluation*, arXiv:1403.7373, 2014.

[2] S.-W. Wang, *A dataset of Sudoku puzzles with difficulty metrics experienced by human players*, IEEE Access **12** (2024), 104254–104262.

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